

Edge Addition Planarity Suite and Generalized Graph Library

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Summary

A *graph* is a mathematical structure comprising a set V of vertices and a set E of edges, with each edge e corresponding to a pair of vertices called the *endpoints* of e . A graph is *planar* if its vertices can be placed in distinct locations on a plane surface and its edges can be drawn on the plane without the edges intersecting, except at their common vertex endpoints. For a planar graph, a planarity algorithm typically outputs a combinatorial data structure that validation code can use to certify the planarity of the input graph. A planar graph drawing is typically produced by a separate algorithm. On the other hand, if an input graph is not planar, then a planarity algorithm typically outputs a minimal subgraph of the input graph that obstructs planarity. Validation code can use a minimal planarity-obstructing subgraph to certify the non-planarity of an input graph. Moreover, a minimal subgraph obstructing planarity can be used to help decide how to amend an input graph to *planarize* it.

Graphs are used to model a very wide array of real-world problems in which there are objects and relationships between the objects. In artificial intelligence, graphs are used to help with reasoning tasks, such as about the relationship pathways between persons of interest in law enforcement or between genes, tissues, diseases, and medications in bioinformatics research. In Physics, graphs and planarity are used to help compute material phase transitions, particle interactions, and electromagnetic duality. Similarly, in chemistry, graphs may be used to represent atoms and their valence bonds in molecules. Planarity testing can help determine feasible molecular arrangements because the molecular graphs for many compounds of interest must be planar. In electronics, a graph may be used to represent the electrical components and wiring in a circuit, such as transistors and etchings on a silicon wafer. The graph of a circuit must be planar or planarized to eliminate short circuits.

The open source software described in this paper provides a highly performant generalized graph library that also implements state-of-the-art algorithms for planarity-related problems. The generalized graph library and planarity-related algorithms are available to C/C++ and Python graph application developers.

Statement of need

The Edge Addition Planarity Suite is an open source software package that was originally developed to implement a new planarity algorithm in (J. M. Boyer & Myrvold, 2004). The Edge Addition Planarity Algorithm is currently regarded as one of two state-of-the-art planarity algorithms due to its conceptual simplicity as well as its speed of execution (Contributors to Wikipedia, 2026). The software package is available from a GitHub repository (J. M. Boyer, 2026) under a 3-clause BSD license.

To enable development of the Edge Addition Planarity Algorithm, as well as the other planarity-related algorithms in (J. M. Boyer, 2006) and (J. M. Boyer, 2012), it was necessary to

41 develop an extensible, generalized graph library. This generalized graph library provides a
42 high-performance solution for the graph algorithm representation and processing needs of a
43 very wide array of graph applications.

44 In addition to enabling graph application development in C/C++, the popularity of Python as
45 a scientific application development language has prompted the need for making this
46 graph library and the algorithms it implements available to Python developers. It can be
47 accessed by a Github repository (J. M. Boyer et al., 2026b) and by simply using `pip install`
48 to get planarity from the Python package installer (J. M. Boyer et al., 2026a).

49 State of the field

50 Several large scientific software packages offer a graph planarity algorithm, which is a complex
51 graph algorithm suitable for benchmarking graph libraries. Mathematica and Wolfram Alpha
52 include planarity algorithms, but these packages are not meant for high performance graph
53 algorithms, so the Library for Efficient Data Types and Algorithms (LEDA) is significantly
54 faster than they are because it is an industrial library designed for computational efficiency.
55 Yet, in (J. M. Boyer et al., 2004), all planarity algorithms implemented in LEDA were shown
56 to be several times slower than the Edge Addition Planarity Algorithm implementation. LEDA
57 also has the disadvantage of being C++ only and has a proprietary license. The only graph
58 library shown (in (J. M. Boyer et al., 2004)) to have similar performance to the Edge Addition
59 Planarity Suite is the Public Implementation of a Graph Algorithm Library and Editor (PIGALE).
60 However, PIGALE is also C++ only, has a GPLv2 license, and lacks advanced algorithms
61 for homeomorphic subgraph search. For example, torus embedding algorithms and torus
62 obstruction isolation algorithms are able to operate differently based on the result of a search
63 for a subgraph homeomorphic to $K_{3,3}$ (Myrvold & Kocay, 2011; Myrvold & Woodcock,
64 2018).

65 Software design

66 The generalized graph library in the Edge Addition Planarity Suite has a carefully curated API
67 and an object-oriented architecture. The base **Graph** class structure includes

- 68 ■ base graph, vertex, and edge structure definitions and getter/setter methods;
- 69 ■ graph, vertex and edge allocation, deallocation, and reset methods;
- 70 ■ methods for iterating through all vertices, edges, and edge adjacency lists;
- 71 ■ graph readers and writers for several formats; and
- 72 ■ a dynamic subclassing/extension mechanisms and virtual function table.

73 The base Graph is extendable with utility methods for exploring a graph, labelling vertices and
74 edges according to a depth-first search (DFS), and for managing connectivity and biconnectivity
75 information. The vertex and edge structures are extended to enable storage of the information
76 generated by the utility methods.

77 A Graph structure extended with the DFS-related utilities is further extendable to a **Planarity**
78 **Graph**, which also further extends the vertex and edge structures. The main additional method
79 provided by a Planarity Graph is `gp_Embed()`, which takes a Planarity Graph as input and
80 outputs either a combinatorial planar embedding or a minimal planarity-obstructing subgraph.

81 Given a Planarity Graph, a number of subclass extensions are available for planarity-related
82 problems. One extension solves outerplanarity. Another implements the planar graph drawing
83 algorithm in (J. M. Boyer, 2006). Three other extensions provide Graph subclasses that solve
84 the homeomorphic subgraph search algorithms appearing in (J. M. Boyer, 2012). All of these
85 extensions further extend the vertex and edge structures and overload the `gp_Embed()` function.

86 In addition to being available to C and C++ developers, the APIs of the Edge Addition

87 Planarity Suite and Generalized Graph Library are also made available via the planarity
88 Python package (J. M. Boyer et al., 2026b). Cython is used to preserve high performance
89 while also broadening availability to the Python developer community.

90 A final aspect of this open source software project is our emphasis on software reliability. We
91 perform validation code on `gp_Embed()` and its five overloads on billions of randomly generated
92 graphs up to 10 million vertices and 30 million edges. The GitHub actions that run on every
93 pull request include sanitizer tests on all graphs on 8 vertices. Finally, the Edge Addition
94 Planarity Suite Testing repository (W. B. K. Boyer & Boyer, 2026) provides our multithreaded
95 Python code for validating `gp_Embed()` and its five overloads on the more than one billion
96 graphs having 11 or fewer vertices, as generated by the geng tool in Nauty and Traces (McKay
97 & Adolfo, 2025).

98 Research impact statement

99 The journal publication of the Edge Addition Planarity Algorithm, (J. M. Boyer & Myrvold,
100 2004), currently has over 360 scholarly citations according to Google Scholar. The algorithm
101 has been implemented in several large open source projects, including Boost, Magma, nauty
102 and Traces (McKay & Adolfo, 2025), and the Open Graph Drawing Framework (web archive).
103 The source code from the Edge Addition Planarity Suite repository has been directly included in
104 several large open source projects, including SageMath, Digraphs, and over 80 Linux platforms
105 releases including for Debian, Devuan, Kali, openSUSE, Raspbian, Ubuntu, Alpine, ALT,
106 Fedora, FreeBSD, Gentoo, Manjaro, MSYS2, and Parabola, according to repology/edge-
107 addition-planarity-suite (web archive) and repology/planarity (web archive).

108 The Python package named planarity was originally developed in 2013 by Aric Hagberg
109 as a way to get the planarity testing and drawing algorithms from (J. M. Boyer & Myrvold,
110 2004) and (J. M. Boyer, 2006) to work with graphs from NetworkX (Hagberg et al., 2008).
111 Recently, Hagberg transferred the planarity GitHub repository (J. M. Boyer et al., 2026b) to
112 the GitHub Graph Algorithms Organization and the planarity Python package (J. M. Boyer
113 et al., 2026a) to the PyPI Graph Algorithms Organization, which are both maintained by the
114 authors. On August 1, 2026, the Top PyPI Packages website indicated that the planarity
115 Python package had over 700,000 downloads for the preceding month and was ranked in the
116 top 1% of PyPI package downloads (ranked 5262 out of 864,036 projects). According to
117 pepy.tech (web archive) the planarity Python package has been downloaded over 2 million
118 times in total. The planarity Python package has now been included as a standard Python
119 package in several versions of Linux including Debian, Devuan, Kali, Raspbian, and Ubuntu,
120 according to repology/python:planarity (web archive).

121 AI usage disclosure

122 No generative AI tools were used in the development of this software, the writing of this
123 manuscript, nor the preparation of supporting materials.

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